

Data Flow Diagram

Date	29 June 2026
Team ID	RW-2026-01
Project Name	Rising Waters
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded Shape	Weather Stations, Government Disaster Management Authority, Citizens, Emergency Response Teams, Administrators	External entities that provide weather and flood-related data or receive flood predictions, alerts, reports, and notifications from the Rising Waters system.
Rectangle with Numbered Header	Collect Weather Data, Preprocess & Validate Data, Flood Prediction Engine, Generate Alerts & Notifications	Processes that collect weather data, preprocess and validate it, predict flood risk using the machine learning model, and generate alerts and notifications.

Rectangle (Solid Fill, No Number)	Weather Database, Historical Flood Database, Machine Learning Model Repository, Alert Logs Database	Data stores used to save weather observations, historical flood records, trained prediction models, and generated alert logs for future analysis and system operations.
Labeled Arrow	Rainfall Data, Temperature Data, Humidity Data, Wind Speed, Water Level, Historical Flood Records, Validated Data, Processed Data, Flood Risk Prediction, Flood Alerts, Safety Instructions, Emergency Notifications, Analytics Reports	Represents the movement of data between external entities, processes, and data stores within the Rising Waters flood prediction system.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

DATA FLOW DIAGRAM – RISING WATERS

